Problem 1

Calculate the time derivative of the following rotation matrices. The angular velocity $\boldsymbol{\omega}$ is expressed in the fixed frame.

(a) $\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ rad/s, $\boldsymbol{R} = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(b) $\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ rad/s, $\boldsymbol{R} = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(c) $\boldsymbol{\omega} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ rad/s, $\boldsymbol{R} = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(d) $\boldsymbol{\omega} = \begin{bmatrix} 0.91340 \\ 0.63240 \\ 0.09754 \end{bmatrix}$ rad/s, $\boldsymbol{R} = \begin{bmatrix} -0.62170 & -0.02073 & -0.78300 \\ -0.13850 & -0.98100 & 0.13590 \\ -0.77090 & 0.19300 & 0.60700 \end{bmatrix}$.

Solution. With the angular velocity expressed in the fixed frame, $\dot{\boldsymbol{R}} = [\boldsymbol{\omega}]_\times \boldsymbol{R}$.

(a) $\dot{\boldsymbol{R}} = \begin{bmatrix} -0.866 & -0.5 & 0 \\ 0.5 & -0.866 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(b) $\dot{\boldsymbol{R}} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -0.5 & 0.866 & 0 \end{bmatrix}$

(c) $\dot{\boldsymbol{R}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0.866 & 0.5 & 0 \end{bmatrix}$

(d) $\dot{\boldsymbol{R}} = \begin{bmatrix} -0.474 & 0.218 & 0.371 \\ 0.643 & -0.178 & -0.631 \\ 0.267 & -0.883 & 0.619 \end{bmatrix}$

Problem 2

A quadcopter is flying around with an inertial measurement unit (IMU) attached 10 cm offset from the center of mass (Figure 1). An external observer measures the position, orientation, linear and angular velocity, and linear acceleration of the quadcopter relative to a fixed ground reference system (origin 0). What is the linear velocity and acceleration, and angular velocity measured by the IMU in the body frame of the quadcopter at this instant? The angular velocity remains constant. *Hint: Do not forget gravity.*

$$ {}^0\boldsymbol{d}_{01} = \begin{bmatrix} 20 \\ 15 \\ 10 \end{bmatrix} \text{m}, \quad {}^0\boldsymbol{R}_1 = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad {}^0\boldsymbol{v}_{01} = \begin{bmatrix} 10 \\ 5 \\ 0.25 \end{bmatrix} \text{m/s}, $$

$$ {}^0\boldsymbol{a}_{01} = \begin{bmatrix} -1 \\ 0 \\ 0.1 \end{bmatrix} \text{m/s}^2, \quad {}^0\boldsymbol{\omega}_{01} = \begin{bmatrix} 0 \\ 0 \\ 2.3 \end{bmatrix} \text{rad/s}. $$

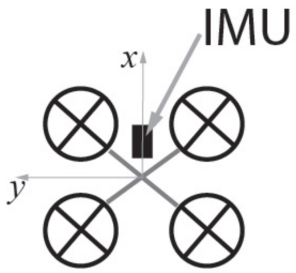


Figure 1: IMU offset from the center of mass.

Solution. The IMU offset in the body frame is ${}^1\boldsymbol{r}_{\text{imu}} = [0.1 \;\; 0 \;\; 0]^\intercal$ m. Since $\boldsymbol{\omega}$ is constant, $\dot{\boldsymbol{\omega}} = \boldsymbol{0}$, so

$$ \boldsymbol{v}_P = \boldsymbol{v}_{\text{com}} + \boldsymbol{\omega} \times \boldsymbol{r}, \qquad \boldsymbol{a}_P = \boldsymbol{a}_{\text{com}} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}). $$

Offset in the fixed frame:

$$ {}^0\boldsymbol{r}_{\text{imu}} = {}^0\boldsymbol{R}_1 \, {}^1\boldsymbol{r}_{\text{imu}} = \begin{bmatrix} 0.05 \\ 0.0866 \\ 0 \end{bmatrix} \text{m}. $$

Velocity:

$$ {}^0\boldsymbol{v}_{\text{imu}} = {}^0\boldsymbol{v}_{01} + {}^0\boldsymbol{\omega}_{01} \times {}^0\boldsymbol{r}_{\text{imu}} = \begin{bmatrix} 9.8008 \\ 5.115 \\ 0.25 \end{bmatrix} \text{m/s}, \qquad {}^1\boldsymbol{v}_{\text{imu}} = {}^0\boldsymbol{R}_1^\intercal \, {}^0\boldsymbol{v}_{\text{imu}}. $$

Acceleration (gravity ${}^0\boldsymbol{g} = [0 \;\; 0 \;\; -9.81]^\intercal \, \text{m/s}^2$ is subtracted, since an IMU measures specific force):

$$ {}^0\boldsymbol{a}_{\text{imu}} = {}^0\boldsymbol{a}_{01} + \left[{}^0\boldsymbol{\omega}_{01}\right]_\times \left[{}^0\boldsymbol{\omega}_{01}\right]_\times {}^0\boldsymbol{r}_{\text{imu}} = \begin{bmatrix} -1.2645 \\ -0.458 \\ 0.1 \end{bmatrix}, \qquad {}^1\boldsymbol{a}_{\text{meas}} = {}^0\boldsymbol{R}_1^\intercal \left({}^0\boldsymbol{a}_{\text{imu}} - {}^0\boldsymbol{g} \right). $$

$$ {}^{1}\boldsymbol{v}_{\text{imu}} = \begin{bmatrix} 9.33 \\ -5.93 \\ 0.25 \end{bmatrix} \text{m/s}, \quad {}^{1}\boldsymbol{a}_{\text{meas}} = \begin{bmatrix} -1.029 \\ 0.866 \\ 9.91 \end{bmatrix} \text{m/s}^2, \quad {}^{1}\boldsymbol{\omega}_{01} = {}^{0}\boldsymbol{R}_1^{\intercal}\, {}^{0}\boldsymbol{\omega}_{01} = \begin{bmatrix} 0 \\ 0 \\ 2.3 \end{bmatrix} \text{rad/s} $$

Problem 3

Consider a three-link manipulator with joint angles $\boldsymbol{\theta} = [\pi/3 \;\; \pi/6 \;\; \pi/6]^{\intercal}$ rad, joint angle rates $\dot{\boldsymbol{\theta}} = [0.5 \;\; -1.0 \;\; 1.5]^{\intercal}$ rad/s, a stationary base, and the following DH table. Assume that the angle rates remain constant. What are the angular velocities $\boldsymbol{\omega}_i$, linear velocities $\dot{\boldsymbol{d}}_{0,i}$, and accelerations $\ddot{\boldsymbol{d}}_{0,i}$ at each frame (i.e., $i = 1, 2, 3$)? In addition, construct the velocity Jacobian. *Hint: The velocity of the end frame is* $\boldsymbol{J}_{\text{v}}\dot{\boldsymbol{\theta}} = [-0.1924 \;\; 0.2498 \;\; -0.3580 \;\; 0.4330 \;\; -0.2500 \;\; 0.5000]^{\intercal}$.

Link	a	d	α	θ
1	0	1	$\pi/2$	$\star$
2	0.5	0	0	$\star$
3	0.3	0	0	$\star$

Solution. Build each link transform from the standard DH form

$$ \boldsymbol{A}_i = \begin{bmatrix} c_\theta & -s_\theta c_\alpha & s_\theta s_\alpha & a\, c_\theta \\ s_\theta & c_\theta c_\alpha & -c_\theta s_\alpha & a\, s_\theta \\ 0 & s_\alpha & c_\alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^{0}\boldsymbol{T}_i = \prod_{k=1}^{i} \boldsymbol{A}_k, $$

and read off the origins $\boldsymbol{d}_{0,i}$ and axes $\boldsymbol{z}_i$ (third column of ${}^{0}\boldsymbol{R}_i$), all in frame $\{0\}$. All joints are revolute and the base is stationary ($\boldsymbol{\omega}_0 = \dot{\boldsymbol{d}}_{0,0} = \boldsymbol{0}$), so with $\boldsymbol{r}_i = \boldsymbol{d}_{0,i} - \boldsymbol{d}_{0,i-1}$ propagate outward for $i = 1, 2, 3$:

$$ \begin{aligned} \boldsymbol{\omega}_i &= \boldsymbol{\omega}_{i-1} + \dot{\theta}_i\, \boldsymbol{z}_{i-1}, \\ \dot{\boldsymbol{\omega}}_i &= \dot{\boldsymbol{\omega}}_{i-1} + \ddot{\theta}_i\, \boldsymbol{z}_{i-1} + \dot{\theta}_i\, \boldsymbol{\omega}_{i-1} \times \boldsymbol{z}_{i-1} \quad (\ddot{\theta}_i = 0), \\ \dot{\boldsymbol{d}}_{0,i} &= \dot{\boldsymbol{d}}_{0,i-1} + \boldsymbol{\omega}_i \times \boldsymbol{r}_i, \\ \ddot{\boldsymbol{d}}_{0,i} &= \ddot{\boldsymbol{d}}_{0,i-1} + \dot{\boldsymbol{\omega}}_i \times \boldsymbol{r}_i + \boldsymbol{\omega}_i \times (\boldsymbol{\omega}_i \times \boldsymbol{r}_i). \end{aligned} $$

The Jacobian columns are

$$ \boldsymbol{J}_i = \begin{bmatrix} \boldsymbol{z}_{i-1} \times (\boldsymbol{d}_{0,3} - \boldsymbol{d}_{0,i-1}) \\ \boldsymbol{z}_{i-1} \end{bmatrix}, \qquad \begin{bmatrix} \dot{\boldsymbol{d}}_{0,3} \\ \boldsymbol{\omega}_3 \end{bmatrix} = \boldsymbol{J}\dot{\boldsymbol{\theta}}. $$

Evaluating in MATLAB (Appendix A):

Frame i	$\boldsymbol{\omega}_i$ (rad/s)	$\dot{\boldsymbol{d}}_{0,i}$ (m/s)	$\ddot{\boldsymbol{d}}_{0,i}$ (m/s^2)
1	[0, 0, 0.5]	[0, 0, 0]	[0, 0, 0]
2	[−0.8660, 0.5, 0.5]	[−0.0625, 0.3248, −0.4330]	[−0.4871, −0.3438, −0.2500]
3	[0.4330, −0.2500, 0.5]	[−0.1924, 0.2498, −0.3580]	[−0.4121, −0.4737, −0.3150]

$$\boldsymbol{J} = \begin{bmatrix} -0.5049 & -0.2549 & -0.1299 \\ 0.2915 & -0.4415 & -0.2250 \\ 0 & 0.5830 & 0.1500 \\ 0 & 0.8660 & 0.8660 \\ 0 & -0.5000 & -0.5000 \\ 1 & 0 & 0 \end{bmatrix}, \qquad \boldsymbol{J\dot{\theta}} = \begin{bmatrix} -0.1924 \\ 0.2498 \\ -0.3580 \\ 0.4330 \\ -0.2500 \\ 0.5000 \end{bmatrix}$$

Appendix A: Code

Problem3.m

Listing 1: Problem3.m

```
clear; clc;

addpath(genpath('../../../Functions'));

%% Joint states and DH parameters
theta    = [pi/3; pi/6; pi/6];
theta_d  = [0.5; -1.0; 1.5];
theta_dd = [0; 0; 0];

a       = [0; 0.5; 0.3];
d       = [1; 0; 0];
alphaDH = [pi/2; 0; 0];

n = numel(a);

%% Frame origins (3 x 4) and z axes (3 x 4), expressed in frame 0
[o, z] = frameData(theta, a, d, alphaDH);

%% Velocity and acceleration recursion (column 1 = stationary base)
% odot is origin
w     = zeros(3, n+1);
odot  = zeros(3, n+1);
alpha = zeros(3, n+1);
oddot = zeros(3, n+1);

for i = 1:n
    r = o(:, i+1) - o(:, i);
    w(:, i+1)     = w(:, i) + theta_d(i) * z(:, i);
    alpha(:, i+1) = alpha(:, i) + theta_dd(i) * z(:, i) ...
                  + theta_d(i) * cross(w(:, i), z(:, i));
    odot(:, i+1)  = odot(:, i) + cross(w(:, i+1), r);
    oddot(:, i+1) = oddot(:, i) + cross(alpha(:, i+1), r) ...
                  + cross(w(:, i+1), cross(w(:, i+1), r));
end

%% Geometric Jacobian in frame 0
J = zeros(6, n);
for i = 1:n
    J(1:3, i) = cross(z(:, i), o(:, end) - o(:, i));
    J(4:6, i) = z(:, i);
end

%% Results (columns 2..4 are frames 1..3)
disp('w ='); disp(w(:, 2:end));
disp('odot ='); disp(odot(:, 2:end));
disp('alpha ='); disp(alpha(:, 2:end));
disp('oddot ='); disp(oddot(:, 2:end));
disp('J ='); disp(J);
```

```
Jtd = J * theta_d;
fprintf('J*theta_d (compare to the problem''s 6-vector):\n'); disp(round(Jtd, 4));

%% Checks
tol = 1e-9;
assert(norm(odot(:, end) - Jtd(1:3)) < tol, 'odot(end) != top half of J*theta_d');
assert(norm(w(:, end)    - Jtd(4:6)) < tol, 'w(end) != bottom half of J*theta_d');

dt = 1e-3;
p = @(t) endOrigin(theta + theta_d*t + 0.5*theta_dd*t^2, a, d, alphaDH);
v_fd   = (p(dt) - p(-dt)) / (2*dt);
acc_fd = (p(dt) - 2*p(0) + p(-dt)) / dt^2;
fprintf('velocity     error: %.3e\n', norm(v_fd   - odot(:, end)));
fprintf('acceleration error: %.3e\n', norm(acc_fd - oddot(:, end)));
assert(norm(v_fd   - odot(:, end))  < 1e-5, 'velocity finite-difference check failed');
assert(norm(acc_fd - oddot(:, end)) < 1e-3, 'acceleration finite-difference check failed');
disp('All checks passed.');

%% Local functions
function [o, z] = frameData(theta, a, d, alphaDH)
    n = numel(a);
    o = zeros(3, n+1);
    z = zeros(3, n+1);
    T = eye(4);
    z(:, 1) = T(1:3, 3);
    for i = 1:n
        T = T * dhTransform(a(i), d(i), alphaDH(i), theta(i));
        o(:, i+1) = T(1:3, 4);
        z(:, i+1) = T(1:3, 3);
    end
end

function p = endOrigin(theta, a, d, alphaDH)
    o = frameData(theta, a, d, alphaDH);
    p = o(:, end);
end
```

Appendix B: Hand Calculations

#1

$$\dot{R} = [\omega_\times] R$$

$$\therefore$$

a) $$\dot{R} = \begin{bmatrix} -.866 & -.5 & 0 \\ .5 & -.866 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

b) $$\dot{R} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -.5 & .866 & 0 \end{bmatrix}$$

c) $$\dot{R} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ .866 & .5 & 0 \end{bmatrix}$$

d) $$\dot{R} = \begin{bmatrix} -.474 & .218 & .371 \\ .643 & -.178 & -.631 \\ .267 & -.883 & .619 \end{bmatrix}$$

#2

$${}^0d_{01} = \begin{bmatrix} 20 \\ 15 \\ 10 \end{bmatrix} m \qquad {}^0R_1 = \begin{bmatrix} .5 & -.866 & 0 \\ .866 & .5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^0v_{01} = \begin{bmatrix} 10 \\ 5 \\ .25 \end{bmatrix} m/s \qquad {}^0a_{01} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} m/s^2$$

$${}^0\omega_{01} = \begin{bmatrix} 0 \\ 0 \\ 2.3 \end{bmatrix} rad/s \qquad {}^1r_{imu} = \begin{bmatrix} .1 \\ 0 \\ 0 \end{bmatrix}$$

f: ${}^1v_{01}$, ${}^1a_{01}$, ${}^1\omega_{01}$

$$v_P = v_{com} + \omega \times r$$

$$a_P = a_{com} + \cancel{\dot{\omega}}^{\,0} \times r + \omega \times (\omega \times r)$$

$${}^0r_{imu} = {}^0R_1\,{}^1r_{imu} = \begin{bmatrix} .05 \\ .0866 \\ 0 \end{bmatrix} m$$

$${}^0v_{imu} = {}^0v_{01} + {}^0\omega_{01} \times {}^0r = \begin{bmatrix} 9.8008 \\ 5.115 \\ .25 \end{bmatrix}$$

$${}^1v_{imu} = {}^0R_1^T \cdot {}^0v_{imu} = \begin{bmatrix} 9.33 \\ -5.93 \\ .25 \end{bmatrix} m/s$$

$${}^0a_{imu} = {}^0a_{com} + [{}^0\omega]_x ([\omega]_x\, {}^0r) = \begin{bmatrix} -1.26 \\ -.458 \\ 1 \end{bmatrix}$$

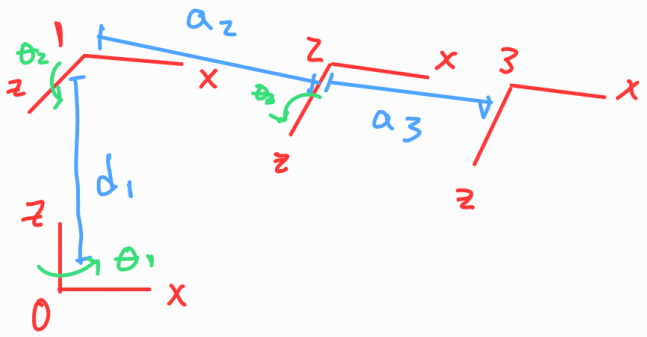
$${}^1a_{meas} = {}^0R_1^T({}^0a_{imu} - {}^0g) = \begin{bmatrix} -1.029 \\ 0.866 \\ 10.81 \end{bmatrix} m/s^2$$

$${}^1\omega_{01} = {}^0R_1^T\,{}^0\omega_{01} = \begin{bmatrix} 0 \\ 0 \\ 2.3 \end{bmatrix} rad/s$$

$$\theta = \begin{bmatrix} \pi/3 & \pi/6 & \pi/6 \end{bmatrix}^T \text{ rad}$$

$$\dot{\theta} = \begin{bmatrix} .5 & -1.0 & 1.5 \end{bmatrix}^T \text{ rad/s}$$

L	a	d	α	θ
1	0	1	$\pi/2$	~~4~~
2	.5	0	0	~~5~~
3	.3	0	0	~~6~~



F: $\omega_i, \dot{d}_{oi}, \ddot{d}_{oi}$; $i = \{1,2,3\}$, $J_v \dot{\theta}$

$$A_i = \begin{bmatrix} C\theta & -S\theta C\alpha & S\theta C\alpha & a \cdot C\theta \\ S\theta & C\theta S\alpha & -C\theta S\alpha & a \cdot S\theta \\ 0 & S\alpha & C\alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_i = \prod_i A_i$$

${}^0T_1 =$ At this point I decided to code the rest